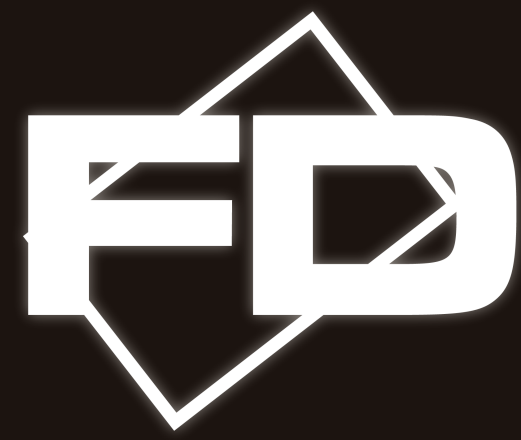




Bias Awareness Card Deck

A structured card-based intervention that prompts cross-functional ML teams to surface bias risks at each stage of the pipeline. Before building, not after deployment.

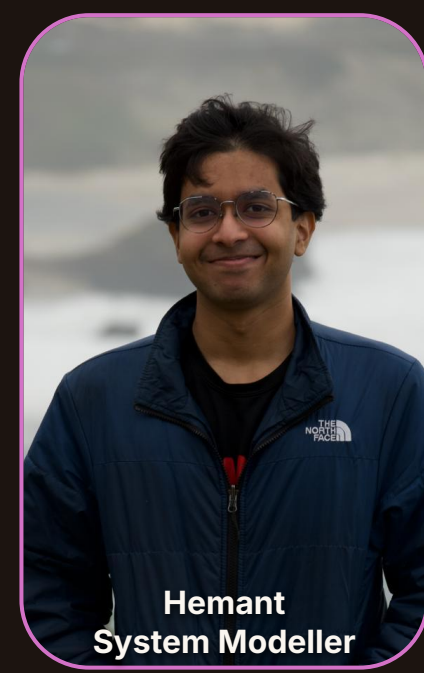
35 cards • 7 stages • 4 types • 15 min • Whole Team

AWARENESS

DISCUSSION

SCENARIO

WILDCARD



Foundation

The Gap

1. Holstein et al. (2019)¹ surveyed 302 ML practitioners and found teams struggled to apply existing fairness auditing methods, especially for bias requiring cross-group comparison.

Industry ML teams care more about fair data collection than algorithmic debiasing, but lack tools for either.

Conclusion: Industry ML practitioners need collaborative, sociotechnical tools for fairness, not just better algorithms, because the biggest barriers they face are organisational and cross-functional, not technical.

2. Madaio et al. (2020)² co-designed a fairness checklist with 48 practitioners and found it only worked when embedded in team workflows and supported by organisational culture.

Fairness checklists only work when embedded in team workflows and backed by organisational culture, not as standalone forms.

Conclusion: Fairness checklists can meaningfully support responsible AI practice, but only when co-designed with teams and embedded in organisational workflows rather than handed down as compliance forms.

3. Deng et al. (2023)³ studied 23 practitioners across 17 companies and found fairness work survives on invisible bridging labour between roles, with no shared artifact built for the job.

AI fairness work survives on invisible bridging labour and piggybacking on privacy processes, where there is no shared artifact built for the job.

Conclusion: Cross-functional collaboration around AI fairness in industry remains fragmented and dependent on invisible bridging labour, pointing to an urgent need for shared artifacts and processes that make fairness a team-wide responsibility.

"Same chart. Same data. Different seeing" - Xavier

COMPAS Statistics

ProPublica's⁴ analysis of Broward County, Florida data found that Black defendants were nearly 2x as likely as white defendants to be wrongly flagged as high-risk by COMPAS, 44.9% vs. 23.5% false positive rate.

Hypotheses

H1: The card deck group detects more bias issues than the control group across all pipeline stages.

H3: The card deck group shows higher inter-rater agreement on severity ratings than control.

H2: The card deck group achieves higher pipeline stage coverage, flagging concerns in more stages than control.

H4: Audit time is slightly higher in the card deck group but remains within the acceptable 45-minute window.

Outcomes

Results Table - Experiment 1 & Experiment 2

Metric	E1 Ctrl	E1 Exp	Δ_1	E2 Ctrl	E2 Exp	Δ_2
Bias severity — overall mean (1-5)	2.77	4.27	+1.50	2.64	4.28	+1.64
Bias severity — S4 Prediction errors	3.17	4.83	+1.66	3.17	4.50	+1.33
Pipeline stage coverage (Q2, stages ticked)	1.83	2.67	+0.84	2.33	2.67	+0.34
Confidence (Q1, 1-5)	2.83	3.50	+0.67	3.00	3.67	+0.67
Audit time (min, fixed)	45	45	—	45	45	—
Sample size (per group)	6	6	—	3	3	—

A 1.5-point gap on a 5-point scale. Same data. Different tool.

Two between-subjects experiments were run (E1 + E2)

- E1: n=12, 6 per group — initial evaluation
- E2: n=6, 3 per group — refined evaluation after a language confound was identified in E1
- E2 instrument redesigned to bilingual per-chart questions to eliminate comprehension as a variable
- Metrics aggregate Part B severity ratings and Part C post-survey responses across both runs

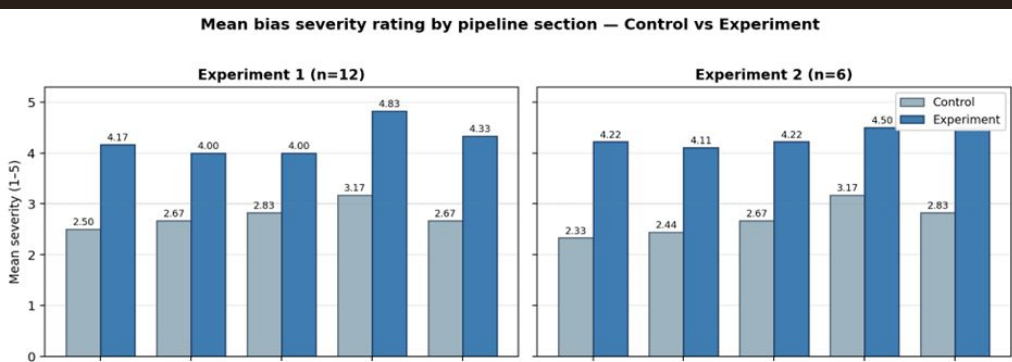


Figure 1: Mean bias severity rating by pipeline section (1-5 scale), Control vs Experiment, across both experimental runs. The experiment group rates severity higher in every section of both runs, with the gap most pronounced in S4 Prediction errors and S5 Deployment.

Findings

What we found

- Card deck group scored consistently higher across both experiments — 4.27 (E1) and 4.28 (E2) vs. control at 2.77 and 2.64
- Gap of +1.5 points on a 5-point scale survived instrument redesign and the E1 language confound fix
- Strongest signal in S4 Prediction errors — experiment group hit 4.83 (E1) and 4.50 (E2), near ceiling
- Deck added most value on non-obvious patterns: sex-based representation, calibration consistency
- Deck added least value where bias was already visually prominent: racial composition, age skew

What this means

- Results support H1 — card deck group detected more bias than control across both runs
- Results partially support H3 — card deck group reported higher audit confidence
- Effect is attributable to the card intervention, not comprehension differences between groups
- Consistent with the deck's design as a cognitive scaffold rather than a checklist
- Caveat: n=18 combined is too small for formal statistical inference — findings are treated as directional DSR evidence, not confirmatory

Severity Rating Tables - Was the data collected significant?

Experiment 1 - Severity Rating Table					
Section	Control avg	Experiment avg	Gap	Mann-Whitney U	Significant
S1	2.50	4.17	+1.67	4.5	Yes*
S2	2.67	4.00	+1.33	4.5	Yes*
S3	2.83	4.00	+1.17	8.0	No
S4	3.17	4.83	+1.66	1.0	Yes*
S5	2.67	4.33	+1.66	1.0	Yes*
Overall	2.77	4.27	+1.50	4.5	Yes*

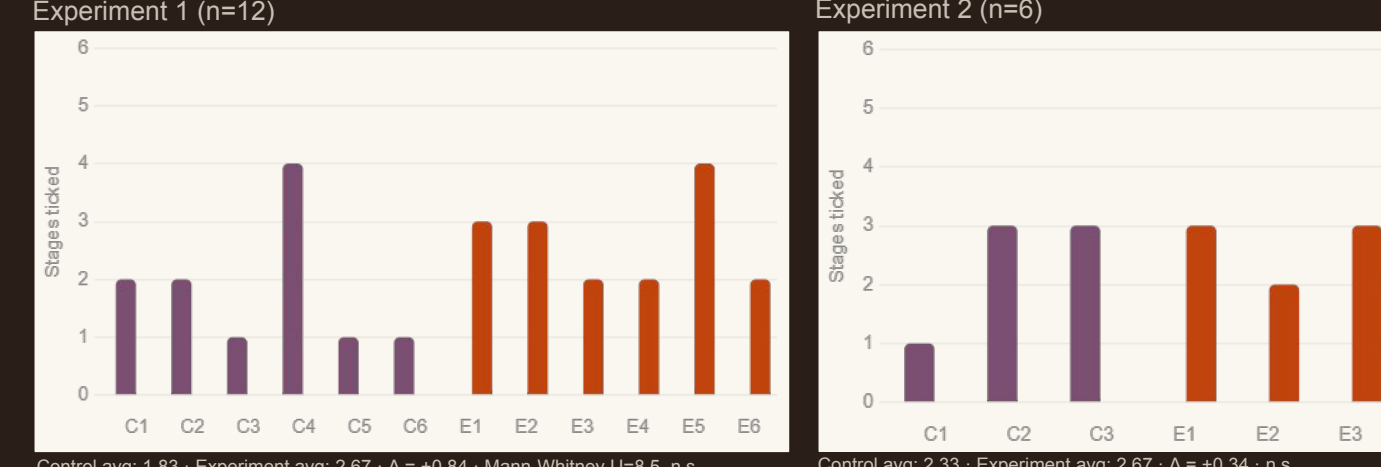
*p < .05 / U < 5

Experiment 2					
Section	Control avg	Experiment avg	Gap	Mann-Whitney U	Significant
S1	2.33	4.22	+1.89	0.0	n.s.*
S2	2.44	4.11	+1.67	0.0	n.s.*
S3	2.67	4.22	+1.55	0.0	n.s.*
S4	3.17	4.50	+1.33	0.0	n.s.*
S5	2.83	4.50	+1.67	0.0	n.s.*
Overall	2.69	4.31	+1.62	0.0	n.s.*

*Direction consistent with H1 - sample too small for formal inference at $\alpha=0.05$

Pipeline Stage Coverage - Stages ticked per participant

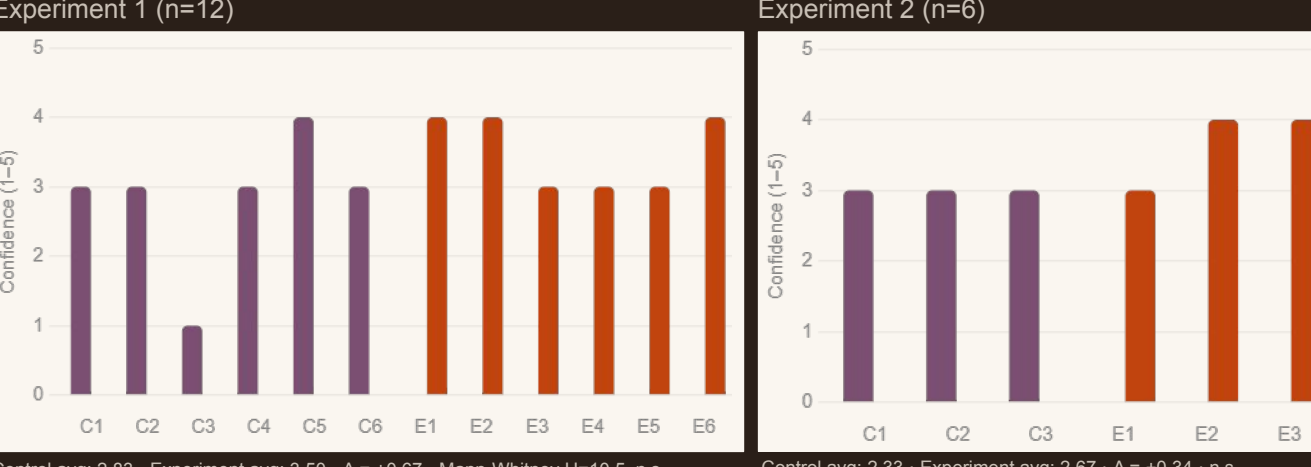
Max 6 stages - Higher = broader pipeline awareness



Direction consistent with H2 - Sample too small for formal inference n.s. = not significant

Confidence Bar Chart - Q1 scores per participant

Q1: "How confident are you that you found all important bias issues?"



Direction consistent with H3 - Notable: C5 gave confidence=4 despite low detection n.s. = not significant

Experiment

Experiment Group

Participants in the experiment condition receive structured, bias-specific prompts before each audit stage:

- Receives a deck of 35 bias awareness cards, each pre-scoped to a specific pipeline stage and a known fairness concern (group representation, label leakage, proxy features, error asymmetry, calibration).
- Refers to the 7 pipeline stages of the COMPAS audit notebook in fixed order, flipping one card per stage.
- Flags bias issues per stage on the response sheet by ticking observed concerns against the card's guiding questions and severity anchor.
- Rates severity of each stage's findings on a 1-5 scale (no concern → severe).
- Completes the post-survey capturing confidence, pipeline stages actively considered, perceived task load, and one open response on what the card surfaced.

Procedure

All participants, regardless of condition, follow the same six-step protocol:

- Briefing (5 min)** — moderator explains the task, opens the Colab notebook, and hands out the response sheet; participants read Part A silently.
- Pre-survey** — participants record baseline confidence and prior exposure to fairness/ML audit concepts before touching the notebook.
- Card session (experiment) / Reflection (control)** — at the start of each of the 7 stages, participants flip the next card and read it silently for 1.5 minutes before looking at any data.

Control Group

Participants in the control condition follow an identical procedure but without bias-specific scaffolding:

- Receives neutral reflection cards with open, non-directive prompts (e.g. "Who is in this dataset?", "What makes a risk score trustworthy?") carrying no fairness vocabulary or severity anchors.
- Follows the same 7-stage structure and the same card-before-cells flow, so exposure order and pacing match the experiment group.
- Refers to the same COMPAS notebook, with identical cells, outputs, and section headings — the only manipulated variable is the card content.
- Operates under the same 45-minute time limit, with moderator calls at 1.5 / 3.5 / 2.0 minutes marks per section to hold timing constant.
- Receives no bias-specific prompts, checklists, or terminology on the reflection cards — fairness language appears only on the shared response sheet, which both groups use.

Participant Response Sheet - Sample Section

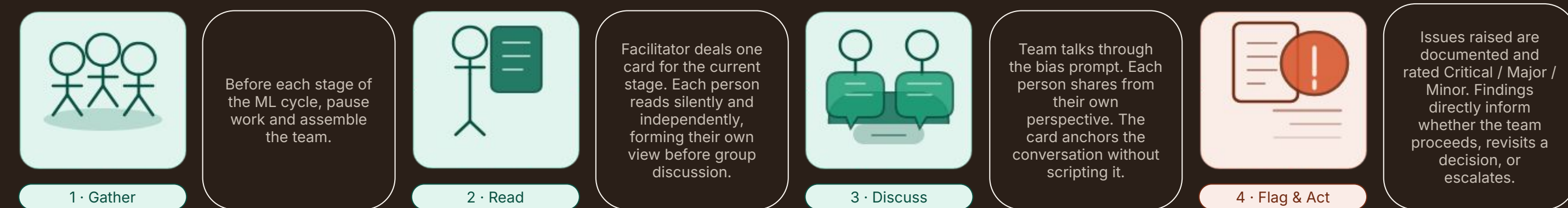
S3 - Section 3: Risk scores across groups		Severity rating	Time: 26 min
第三節：各群體的风险评分		严重程度	预计时间
Feature Engineering 特征工程			
Which racial group is rated medium/high risk most often? Compare to the group rated least often.			
Chart A	图表 A	Circle one: 圈出:	
哪个种族群体被评价为中高风险的频率最高？与频率最低群体相比如何？		1 2 3 4 5	
		No concern 无关注	Moderate 中等
		Severe 严重	
Is the average risk score meaningfully different across racial groups?		Circle one: 圈出:	
Chart B	图表 B	1 2 3 4 5	
不同种族群体的平均风险评分是否存在显著差异？		No concern 无关注	Moderate 中等
		Severe 严重	
Is the sex difference in high-risk rates larger or smaller than the racial differences in Chart A?		Circle one: 圈出:	
Chart C	图表 C	1 2 3 4 5	
性别间的高风险评分差异是否大于或小于图表 A 中的种族差异？		No concern 无关注	Moderate 中等
		Severe 严重	

- Each section corresponds to one of the 5 stages of the COMPAS notebook audit.
- Contains per-chart questions asking participants to identify bias patterns in the data they just viewed
- Participants circle a severity rating from 1 (no concern) to 5 (severe) for each chart
- Ratings create a quantitative record of how much bias was detected and how seriously it was judged
- Bilingual English-Chinese format was introduced in Experiment 2 after a language confound was identified in Experiment 1
- In E1, some participants may have misread questions due to unfamiliarity with fairness terminology in English
- The redesigned bilingual sheet ensured comprehension was not a barrier to detecting bias
- This isolated the card deck as the sole manipulated variable between the two groups

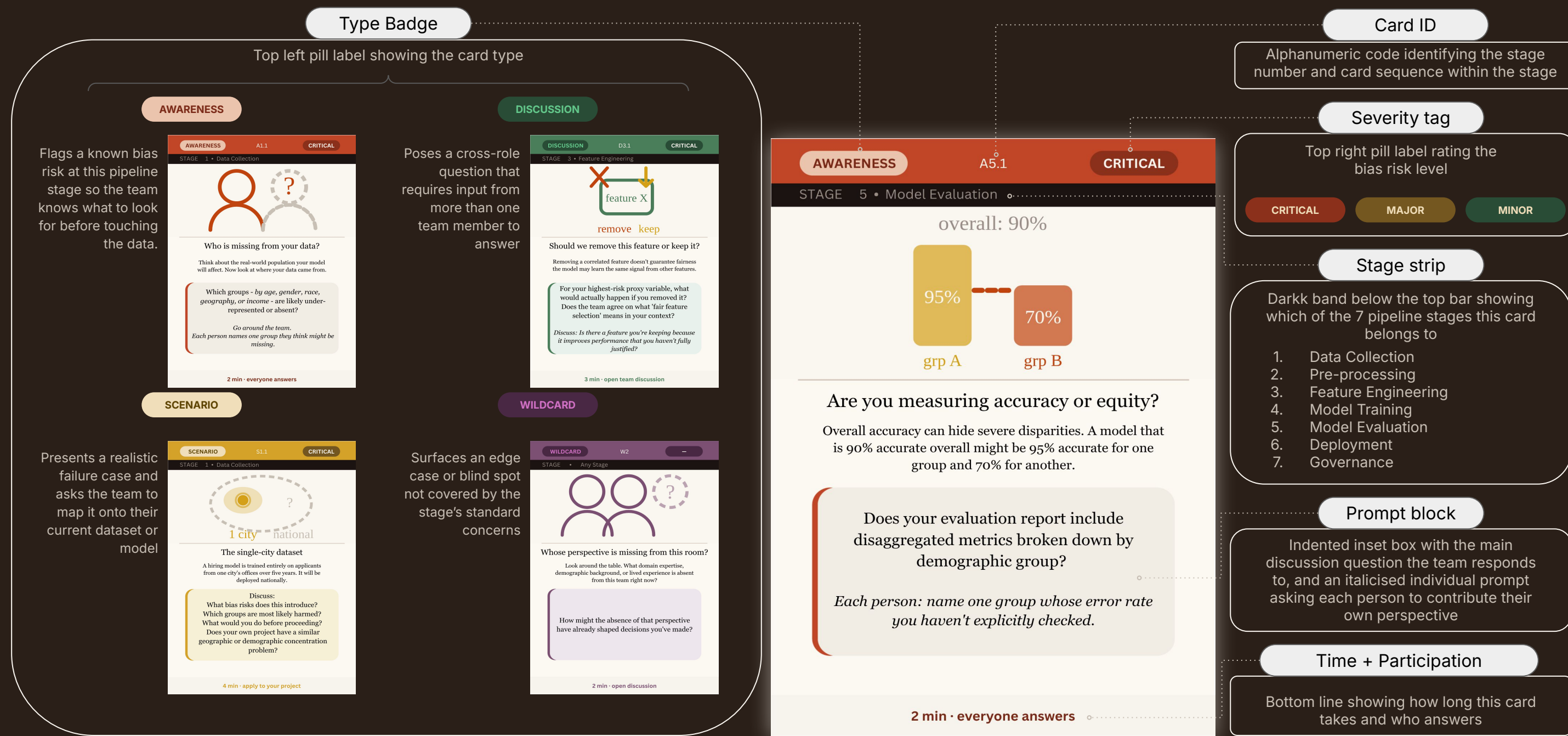
Methodology

35 cards • 7 stages • 4 types • 15 min • Whole Team.

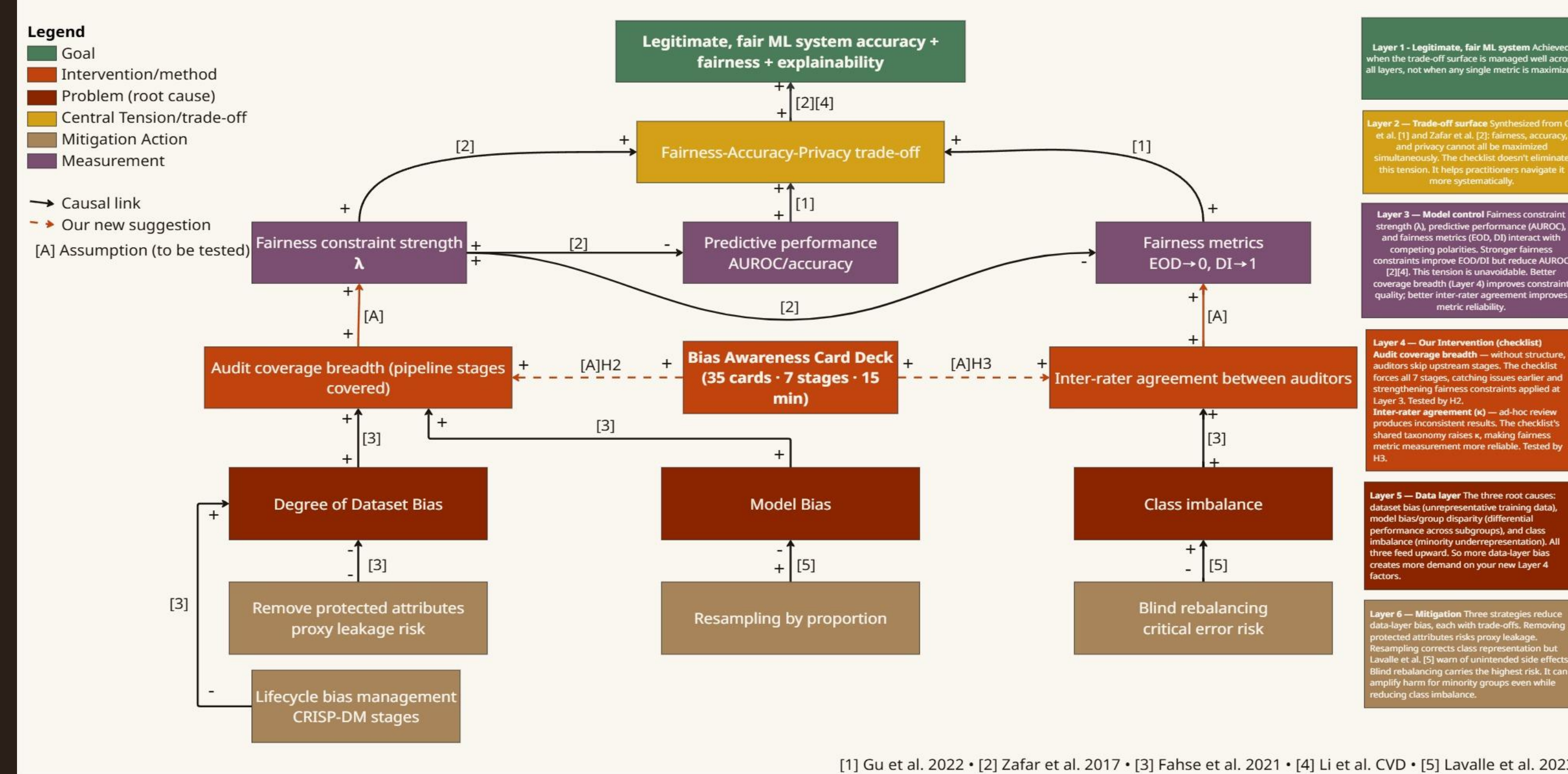
4-Step Workflow



Card Deck Specifications



Reference Model with Proposed Method



The intended effect of our intervention: Card deck raises bias awareness → teams catch more issues across pipeline stages → fairer model outputs. [A] nodes are assumptions pending experimental confirmation

- Yu, Xin, Yan Fang, and Jian Liang. "Privacy-Accuracy-Fairness Trade-offs in Federated Learning." *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 4, 2022, pp. 1-10.
- Zafar, Muhammad Bilal, Isabel Valera, Manuel Gomez Rodriguez, and Krishna P. Gummadi. "Fairness Constraints: Mechanisms for Fair Classification." *Proceedings of the 2018 International Conference on Artificial Intelligence and Statistics*, PMLR, 2017, pp. 962-970. <https://doi.org/10.1145/3038392.3052860>
- Madaio, Michael A., Luke Stark, Jennifer Wortman Vaughan, and Hanna Wallach. "Co-Designing Checklists to Understand Organizational Challenges and Opportunities around Fairness in AI." *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, ACM, 2020, pp. 1-14. <https://doi.org/10.1145/3376445>
- Deng, Wesley Hanwen, Bhavya Bhavya, Kenneth Holstein, and Haiyi Zhu. "Investigating Practices and Opportunities for Cross-Functional Collaboration around AI Fairness in Industry." *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, ACM, 2023, pp. 806-816. <https://doi.org/10.1145/3593073.3594937>
- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine Bias. *ProPublica*. <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- Gu, Xin, Yan Fang, and Jian Liang. "Privacy-Accuracy-Fairness Trade-offs in Federated Learning." *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 4, 2022, pp. 1-12.
- Zafar, Muhammad Bilal, Isabel Valera, Manuel Gomez Rodriguez, and Krishna P. Gummadi. "Fairness Constraints: Mechanisms for Fair Classification." *Proceedings of the 2018 International Conference on Artificial Intelligence and Statistics*, PMLR, 2017, pp. 962-970. <https://doi.org/10.1145/3038392.3052860>
- Fahse, Tobias, Christoph Ihl, and Vera Wedde. "Managing Bias in Machine Learning Projects." *Proceedings of the 54th Hawaii International Conference on System Sciences*, 2021, pp. 1-10.
- Li, Xiao Hui, Chui Chui Foy, Keem Shah Xu, and Kay Chen Tan. "Evaluating and Mitigating Bias in Machine Learning Models for Cardiovascular Disease Prediction." *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 1, 2022, pp. 1-12.
- Lavalle, Maria, Alejandro Maté, and Juan Trujillo. "Bias and Rebalancing in Machine Learning: Unintended Side Effects and Mitigation Strategies." *Information Systems*, vol. 118, 2023, pp. 1-15.

Reflections

Small sample. Strong signal. A larger study is the next step.

Impact & Contributions

Impact

The Bias Awareness Card Deck addresses a gap identified across three independent bodies of literature: Holstein et al. (2019) found practitioners lack collaborative fairness tools, Madaio et al. (2020) found checklists only work when embedded in team workflows, and Deng et al. (2023) found no shared artifact exists for cross-functional fairness work.

Contributions

Fairdeck is that artifact. By structuring a 15-minute team session before each pipeline stage, it transforms bias awareness from an individual afterthought into a shared, repeatable team practice, without requiring technical expertise or adding significant time overhead.

Limitations

- n=18 combined across both experiments is too small for formal statistical inference, findings are treated as directional DSR evidence rather than confirmatory results
- Participant pool limited to SUTD students, not industry practitioners, limiting ecological validity
- Single dataset (COMPAS) results may not generalise to other domains or bias types beyond criminal justice redivism scoring
- Individual silent reflection rather than real cross-functional team context, the card deck is designed for whole teams, but the experiment necessarily tested individuals to preserve individual-level data at small sample sizes

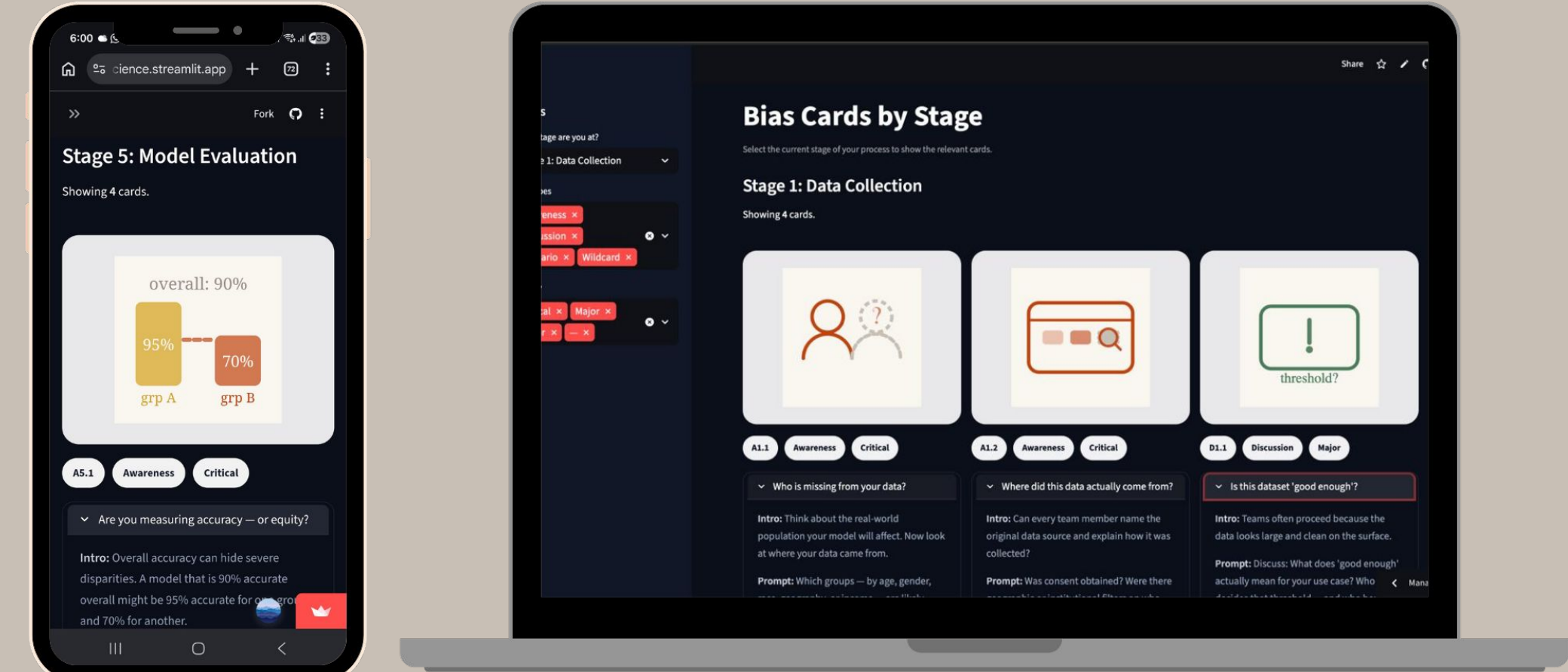
Tools & Method Refinements

- A language confound was identified in Experiment 1 — control group participants with limited English proficiency produced score patterns consistent with comprehension failure rather than genuine low bias detection. The instrument was redesigned for Experiment 2 with bilingual English-Chinese per-chart questions, isolating the card deck as the sole manipulated variable
- Participant E4 suggested adding error bars and brief chart introductions before each section, future notebook versions should include these to reduce cognitive load
- Participant E5 noted certain bias patterns were visually obvious without the card, future card design should focus prompts more heavily on non-obvious patterns where the deck adds the most unique value
- The physical card deck could be complemented by a digital version integrated directly into ML workflow tools, surfacing relevant cards automatically at each pipeline stage

Future Directions

- Run a larger study with genuine cross-functional industry teams — data scientists, product managers, domain experts — to test the card deck in its intended context
- Extend to multiple datasets across different high-stakes domains: healthcare, hiring, credit scoring
- Explore a digital version of the card deck integrated into ML workflow tools such as Jupyter notebooks or MLflow, surfacing the relevant card automatically at each pipeline stage
- Conduct a longitudinal study to test whether teams who use the card deck during development produce models with measurably better fairness metrics at deployment

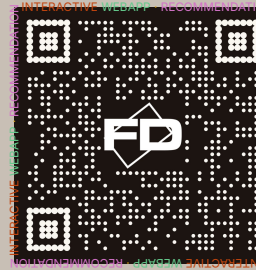
Browse the Deck - 35 Cards, Any Stage, Any Type



The webapp grew out of a practical gap the experiment exposed: exhibition visitors cannot interact with 35 physical cards on a poster.

After running two controlled experiments, it became clear that the deck's value lies in being able to browse and filter cards in context and that needed to be accessible to anyone at the booth without a facilitator or a physical copy.

Scan the QR code to filter all 35 cards by pipeline stage or card type.



Bibliography

- Holstein, Kenneth, Jennifer Wortman Vaughan, Hal Daumé II, Miro Dudík, and Hanna Wallach. "Improving Fairness in Machine Learning Systems: What Do Industry Practitioners Need?" *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, ACM, 2019, pp. 1-16. <https://doi.org/10.1145/3300605.3300723>
- Madaio, Michael A., Luke Stark, Jennifer Wortman Vaughan, and Hanna Wallach. "Co-Designing Checklists to Understand Organizational Challenges and Opportunities around Fairness in AI." *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, ACM, 2020, pp. 1-14. <https://doi.org/10.1145/3376445>
- Deng, Wesley Hanwen, Bhavya Bhavya, Kenneth Holstein, and Haiyi Zhu. "Investigating Practices and Opportunities for Cross-Functional Collaboration around AI Fairness in Industry." *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, ACM, 2023, pp. 806-816. <https://doi.org/10.1145/3593073.3594937>
- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine Bias. *ProPublica*. <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- Gu, Xin, Yan Fang, and Jian Liang. "Privacy-Accuracy-Fairness Trade-offs in Federated Learning." *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 4, 2022, pp. 1-12.
- Zafar, Muhammad Bilal, Isabel Valera, Manuel Gomez Rodriguez, and Krishna P. Gummadi. "Fairness Constraints: Mechanisms for Fair Classification." *Proceedings of the 2018 International Conference on Artificial Intelligence and Statistics*, PMLR, 2017, pp. 962-970. <https://doi.org/10.1145/3038392.3052860>
- Fahse, Tobias, Christoph Ihl, and Vera Wedde. "Managing Bias in Machine Learning Projects." *Proceedings of the 54th Hawaii International Conference on System Sciences*, 2021, pp. 1-10.
- Li, Xiao Hui, Chui Chui Foy, Keem Shah Xu, and Kay Chen Tan. "Evaluating and Mitigating Bias in Machine Learning Models for Cardiovascular Disease Prediction." *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 1, 2022, pp. 1-12.
- Lavalle, Maria, Alejandro Maté, and Juan Trujillo. "Bias and Rebalancing in Machine Learning: Unintended Side Effects and Mitigation Strategies." *Information Systems*, vol. 118, 2023, pp. 1-15.